GRADUATION PROJECT DEFENSE

Syrian Private University | SPU



Autonomous Vision-Guided Robotic Manipulation System for Intelligent Jigsaw Puzzle Assembly

A 4-DOF Robotic Arm System Integrated with Eye-in-Hand Computer Vision, ROS 2 Humble & Gazebo Simulation.

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Project Scope & Domain

Autonomous Vision-Guided 4-DOF Robotic Manipulation System is an integrated robotics and computer vision platform designed for intelligent 2D planar jigsaw puzzle assembly. The system utilizes an optimized 4-DOF robotic manipulator structure to execute precise pick-and-place tasks with minimized computational overhead. It incorporates an Eye-in-Hand camera setup using OpenCV for real-time contour extraction, centroid pixel detection, and 3D workspace coordinate mapping (X, Y, Z, θ). The entire physical dynamics, environment spawning, collision avoidance, and joint trajectory planning are developed and verified within ROS 2 Humble, Gazebo, and MoveIt 2 simulation frameworks. Physical hardware deployment, dynamic 3D spatial assembly beyond 2D planes, and deep-learning-based piece generation are outside the scope of this current version.

COMPARATIVE ANALYSIS

Literature Review & Benchmarking

Reference Paper	Methodology	Strengths	Limitations	Our Advantage
Felzenszwalb & Huttenlocher (2004)	Graph-based image segmentation.	Strong theoretical solver.	No physical or robotic manipulation integration.	Full physical simulation & robotic execution in ROS 2.
Guo et al. (2018)	Eye-in-Hand vision with 6-DOF arm.	High precision assembly.	6-DOF is computationally heavy & over-engineered for 2D.	Optimized 4-DOF system tailored for planar puzzle tasks.
Sharma et al. (2023)	4-DOF arm with Open-Loop vision.	Simple 4-DOF hardware structure.	Open-loop control lacks dynamic error correction.	MoveIt 2 active feedback with collision-free path planning.
Our System (SPU 2026)	**Closed-Loop Eye-in-Hand + 4-DOF IK**	**Low-cost, high precision.**	-	**Integrated ROS 2, MoveIt 2 & Gazebo Pipeline.**

Key Contributions

Main Bullet Points:

1- From Abstract Image Matching to Physical Simulation

Bridges the gap between theoretical 2D image matching (Felzenszwalb & Huttenlocher, 2004) and physical execution by deploying a 4-DOF manipulator in Gazebo with realistic physics (gravity, friction, and collisions).

2- Next-Gen Middleware & Dynamic Trajectory Planning

Replaces legacy, static frameworks (Guo et al., 2018) with **ROS 2 Humble** and **MoveIt 2**, enabling real-time adaptive motion planning and dynamic object spawning.

3- Closed-Loop Eye-in-Hand Visual Feedback

Advances beyond open-loop pick-and-place systems (Sharma et al., 2023) by implementing continuous Eye-in-Hand visual feedback via OpenCV for real-time feature detection and 3D pose estimation (X, Y, Z, θ).



KEY MILESTONES

Core Project Objectives

Main Objectives

Cost-Effective 4-DOF Architecture

Eliminate unnecessary degrees of freedom for planar assembly to lower hardware overhead.

Precise Spatial Mapping

Transform camera pixel frame coordinates into 3D Gazebo world coordinates seamless execution.

Autonomous Visual Pipeline

Extract puzzle contours, centroid pixel coordinates, and orientation via OpenCV.

MoveIt 2 Trajectory Planning

Generate collision-free smooth trajectories using MoveIt 2 Inverse Kinematics solvers.

Secondary Objectives

3D Robotic Modeling & Vision Integration

Build and develop a 3D robot model in URDF format with a virtual camera mounted in an Eye-in-Hand configuration.

Computer Vision & Feature Extraction

Process images in real-time to extract contours and determine precise pixel coordinates for the puzzle pieces.

ROS 2 Software Architecture & Communication

Develop software nodes in ROS 2 (Nodes & Topics) to seamlessly publish and subscribe to coordinate data.

Autonomous Pick-and-Place Execution

Execute automated pick-and-place and puzzle assembly operations fully without human intervention.

Technical Specifications & System Constraints

Content:

Physical & Workspace Limits: 4-DOF manipulator structure with a maximum reach of 40 cm in 3D workspace.

Kinematic Performance: Linear speed ≥ 0.2 m/s, angular velocity ≥ 1.5 rad/s, and positioning accuracy within ± 2 mm.

Vision & Sensing: Eye-in-Hand camera with a 60° FOV, measuring position and orientation at a frame rate ≥ 30 FPS.

Control & Computation: ROS 2 node processing frequency ≥ 30 Hzwith end-to-end response time < 50 ms.



EXECUTION FLOW

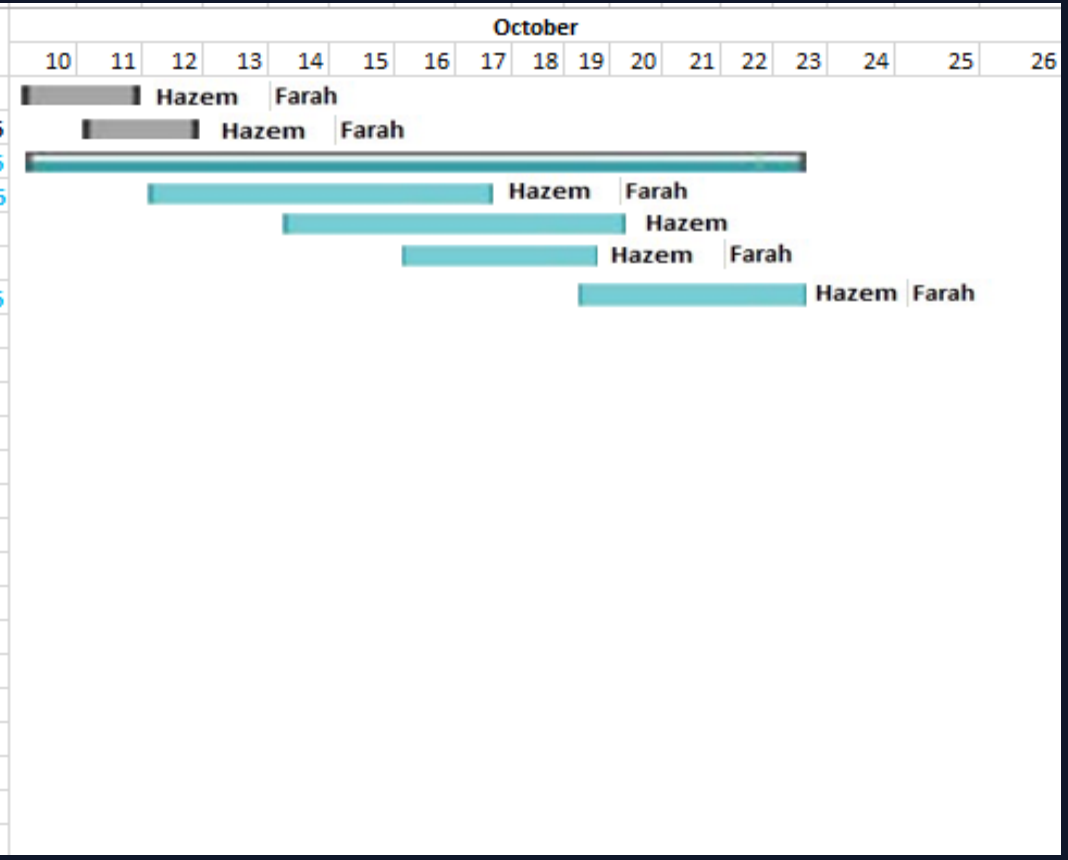
Development Methodology To Simulation



PROJECT PLAN

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Task Name	Duration	Start	Finish
Initial scope determination	2 Days	Sat 10/10/2026	Sun 11/10/2026
Development Environment & ROS 2 Workspace Setup	2 Days	Sun 11/10/2026	Mon 12/10/2026
Sprint 1: (URDF & Simulation Setup)	16 Days	Mon 12/10/2026	Mon 23/10/2026
Literature Review	3 Days	Mon 12/10/2026	Wed 14/10/2026
4-DOF Mechanical Design & Kinematics	4 Days	Sat 14/10/2026	Tue 17/10/2026
URDF Model Development (Links, Joints, & Inertia Properties)	4 Days	Mon 16/10/2026	Thu 19/10/2026
Gazebo Physics World Integration & Testing	5 Days	Thu 19/10/2026	Mon 23/10/2026



NEXT STEPS

Upcoming Development Milestones

MoveIt Setup Assistant

Finalize the arm MoveIt configuration package to generate kinematic groups and end-effector definitions.

Inverse Kinematics Integration

Implement precise spatial coordinate targeting using MoveIt 2 IK solvers for smooth motion trajectory.

Gripper Mechanism Activation

Integrate a vacuum gripper plugin in Gazebo for physical attachment and placement of puzzle pieces.

Full End-to-End Pipeline

Execute complete automated loop: Detection → MoveIt Planning → Gripping → Assembly Placement.

Thank You!

Questions & Academic Discussion

Autonomous Vision-Guided Robotic Manipulation System

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